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Empirical Article

Two systems, two timelines: Computational evidence for dissociable development in inhibitory control across childhood and adolescence

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Abstract

This study examined inhibitory control development across two samples of Chinese children: a cross-sectional ($n = 1{,}122$; 45.5% female; 91.9% Han; M_{age}=12.42 years, range: 6.0–18.7 years) with 6-month longitudinal follow-up and independent validation sample ($n = 1{,}026$; 45.1% female; 90.8% Han; M_{age}=12.44 years, range: 6.1–18.8 years). Generalized additive models applied to *Stroop* and *Go/No-Go* tasks revealed four-phase nonlinear developmental trajectories. Response inhibition stabilized by 13.4 years, while interference inhibition developed until 15.8 years. Hierarchical drift diffusion modeling indicated that interference inhibition development was associated with enhanced information accumulation (drift rate), whereas response inhibition development was associated with response bias adjustment (starting point), with shared processing speed improvements—a dissociation emerging across development. These findings support a decision-computational framework for understanding inhibitory control development.

Keywords inhibitory control, nonlinear development, drift diffusion model, generalized additive model, machine learning

Lay summary

The underlying computational mechanisms driving the development of children's ability to control impulses and ignore distractions remain unclear. We studied 2,148 Chinese children aged 6–18 years using tasks measuring two types of control: ignoring conflicting information and stopping automatic responses. Both abilities improved through four stages, with the quickest gains occurring between ages 6 and 8 years. Success in stopping responses stabilized by age 13 years, while success in ignoring conflicting information developed until age 16 years. Per computational modeling, these skills rely on different mental processes—one reflects improved efficiency in processing conflicting information; the other reflects adjustments in response tendencies. Younger children use general strategies for both skills, but, by late adolescence, specialized processes develop for each.

Introduction

A child waiting at a crosswalk sees the pedestrian signal turn red just as the adjacent traffic light turns green, and steps forward anyway, misled by the conflicting cue. Another child, walking home from school, spots a friend waving from across a busy street and impulsively darts toward them; the urge to respond overriding any deliberation. Both errors reflect failures of inhibitory control, yet they may arise from different demands: the first requires filtering out misleading perceptual information, while the second primarily requires suppressing an automatic motor response triggered by a salient external stimulus. Whether these demands draw on a common underlying capacity or distinct mechanisms is a question that has occupied researchers for decades, and one with direct implications for understanding and supporting children's inhibitory development.

Inhibitory control—the capacity to suppress inappropriate responses in favor of goal-directed behavior (Zhou et al., 2012)—develops gradually from early childhood through adolescence (Constantinidis & Luna, 2019; Durston et al., 2002) and predicts numerous positive developmental outcomes, including enhanced social competence (Beauchamp & Anderson, 2010), improved academic achievement (Allan et al., 2014; Valiente et al., 2010), and fewer behavioral and emotional problems (Eisenberg, Spinrad, & Eggum, 2010; Moffitt et al., 2011). Conversely, deficits in inhibitory control are implicated in attention difficulties, emotion dysregulation, and impulse control challenges across development (Nigg, 2017; Rothbart & Bates, 2006). How researchers

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conceptualize and measure inhibitory control, however, varies across theoretical traditions. The executive function (EF) approach, rooted in cognitive neuroscience, emphasizes performance-based paradigms and conceptualizes inhibitory control as a core cognitive process (Diamond, 2013; Fiske & Holmboe, 2019; Miyake et al., 2000). In contrast, the effortful control approach, grounded in temperament research, frames this capacity as a temperamental dimension typically assessed through questionnaires (Rothbart et al., 2003). Despite terminological distinctions, both traditions implicate prefrontal–cingulate maturation as the neural foundation for developmental improvements (Luna et al., 2015; Kim-Spoon et al., 2019; Tiego et al., 2020; see Table S1 for a detailed comparison). Critically, questionnaire-based effortful control assessments capture stable individual differences but cannot reveal the cognitive processes underlying task performance (Rothbart & Posner, 2022), whereas performance-based EF paradigms generate trial-level data amenable to computational decomposition (Ambrosi et al., 2020; Ratcliff et al., 2012; Rey-Mermet et al., 2018). This capacity for fine-grained process analysis renders the EF approach particularly suited for addressing a central unresolved question: whether inhibitory control operates as a unitary capacity or comprises functionally distinct subcomponents.

This unity-versus-diversity debate remains contested (Palmer et al., 2025): some evidence supports common underlying mechanisms across inhibitory domains (Wessel & Anderson, 2024), whereas other research indicates that inhibitory components do not consistently load onto a single factor (Hartung et al., 2020; Shields & Yonelinas, 2025) and may follow divergent developmental trajectories (Cragg, 2016). If inhibitory control operates as a unified capacity, children should perform consistently across tasks, and computational models should reveal common latent processes; if it comprises distinct subcomponents, performance and model parameters should dissociate across domains. Here, we examine whether interference inhibition and response inhibition follow independent developmental trajectories and rely on distinct computational mechanisms. To address this question, we integrate flexible nonlinear modeling of behavioral trajectories with computational decomposition of underlying decision processes.

The multicomponent nature of inhibitory control

Within the EF framework, inhibitory control encompasses multiple components that vary in their cognitive demands and neural substrates, including cognitive inhibition (suppressing irrelevant thoughts or memories; Friedman & Miyake, 2004), behavioral inhibition (withholding responses in novel contexts; Nigg, 2000), and oculomotor inhibition (suppressing reflexive eye movements; Roberts et al., 2011). Among these, two components have received the most robust empirical support: interference inhibition, which resolves conflict between competing stimulus dimensions, and response inhibition, which suppresses prepotent motor responses (Nigg, 2000; Palmer et al., 2025). Factor analytic studies consistently identify these as separable latent constructs (Friedman & Miyake, 2004; Shields & Yonelinas, 2025), and neuroimaging research reveals distinct neural substrates—interference inhibition engages the anterior cingulate cortex and dorsolateral prefrontal cortex for conflict detection and resolution (Botvinick et al., 2001; Isherwood et al., 2023), whereas response inhibition primarily recruits the right inferior frontal gyrus and presupplementary motor area for motor suppression (Aron et al., 2014; Fiske & Holmboe, 2019; Isherwood et al., 2023). Nevertheless, recent within-subject neuroimaging also reveals substantial overlap in frontoparietal activation across both task types (Palmer et al., 2025), suggesting that complete dissociation may be an oversimplification.

Developmental patterns of interference and response inhibition

Inhibitory control emerges early in life and undergoes substantial development prior to school age. Rudimentary forms of inhibitory capacity appear during the first year, with infants demonstrating basic response suppression by 12 months (Diamond, 1990). The toddler and preschool years are characterized by particularly rapid gains, as children become increasingly capable of delaying gratification, inhibiting prepotent responses, and resolving simple conflicts (Carlson et al., 2005; Garon et al., 2008; Kochanska et al., 2000). By age 4–5 years, children can successfully perform simplified inhibitory paradigms such as the *day–night* task (Gerstadt et al., 1994) and the *Head-Toes-Knees-Shoulders* task (Ponitz et al., 2009), marking an important developmental milestone. Despite these shared early foundations, interference and response inhibition appear to follow distinct developmental timetables during childhood and adolescence. Relative to interference inhibition, response inhibition demonstrates earlier competence, with rapid gains extending through middle childhood (Cohen et al., 2016; Constantinidis & Luna, 2019). Interference inhibition, by contrast, shows more protracted development, with marked improvements between ages 7 and 10 years and continued refinement into adolescence (Luna et al., 2015). These behavioral patterns broadly parallel distinct trajectories of neural maturation: response inhibition improvements are more closely associated with frontostriatal circuit development, whereas interference inhibition gains are more strongly linked to frontoparietal network maturation—though considerable overlap exists (Constantinidis & Luna, 2019; Crone & Steinbeis, 2017; Madsen et al., 2010).

However, existing studies have yielded inconsistent findings regarding developmental timing, critical periods, and growth profiles—discrepancies that likely reflect methodological limitations. Previous research has either imposed predetermined growth models (e.g., linear or quadratic functions) or relied on cross-sectional age group comparisons, both of which may obscure nonlinear dynamics (see Table S2). Addressing these limitations requires flexible analytical frameworks that do not impose predetermined functional forms. Generalized additive models (GAMs) with derivative analysis offer a promising solution by capturing nonlinear dynamics and identifying periods of developmental acceleration and deceleration through first derivatives (Hao et al., 2025; Tervo-Clemmens et al., 2023)—patterns that traditional methods may fail to detect—yet systematic comparisons of interference and response inhibition trajectories using such methods remain scarce. This gap is particularly notable for the developmental window from ~6 years of age—when children first demonstrate reliable performance on computerized inhibitory paradigms

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(Brocki & Bohlin, 2004; Davidson et al., 2006) and when EF begins differentiating into separable components (Lee et al., 2013; Shing et al., 2010)—through late adolescence around age 18 years, when inhibitory performance approaches adult-like levels (Luna et al., 2015; Ordaz et al., 2013) and prefrontal maturation nears completion (Blakemore & Choudhury, 2006; Gogtay et al., 2004).

Computational approaches to understanding inhibitory development

While conventional behavioral measures have provided foundational insights, they face significant limitations in characterizing the complex developmental dynamics of inhibitory control. Traditional paradigms quantify inhibitory capacity by comparing performance between conflict and baseline conditions, primarily through reaction time differences (interference effects) and accuracy rates. This approach, however, conflates multiple cognitive processes within single behavioral measures, potentially obscuring the precise mechanisms driving developmental change (Hedge et al., 2018; Rey-Mermet et al., 2018). For instance, identical interference effects may reflect either slow overall processing with intact conflict resolution or rapid processing with specific inhibitory deficits—with different theoretical and practical implications.

Computational cognitive modeling offers a powerful solution by decomposing complex behavior into theoretically grounded component processes (Forstmann et al., 2016; Isherwood et al., 2023; Wilson & Collins, 2019). Rather than treating reaction times and accuracy as separate, isolated measures, computational models integrate multiple behavioral indicators within unified theoretical frameworks. The hierarchical drift diffusion model (HDDM), grounded in evidence accumulation theory, exemplifies this approach by parsing decision-making into distinct cognitive operations (Forstmann et al., 2016; Isherwood et al., 2023). The model characterizes decisions as continuous evidence accumulation toward response boundaries, with four core parameters (Wiecki et al., 2013)—drift rate (v), decision boundary (a), starting point bias (z), and nondecision time (t)—mapping onto distinct cognitive operations (see Table S3 for detailed descriptions).

This framework generates specific predictions about inhibitory development. The conflict monitoring hypothesis (Botvinick et al., 2001) suggests that interference inhibition depends critically on detecting and resolving conflict, which should manifest computationally as age-related improvements in drift rate as conflict resolution becomes more efficient. Response inhibition, conversely, may depend more on strategically elevating decision boundaries (proactive control) or adjusting starting points away from prepotent response tendencies. Recent developmental applications support these predictions. Ambrosi et al. (2019) applied a drift diffusion model (DDM) to conflict task performance in 5- to 6-year-old children, revealing that these children's prolonged reaction times reflected notably slower drift rates—alongside elevated decision thresholds and longer nondecision times—compared with adults. In contrast, Ratcliff et al. (2012) applied diffusion modeling to children's (ages 8–15 years) two-choice decision-making (numerosity discrimination, lexical decision), revealing that age-related improvements involved narrowing decision boundaries and reduced nondecision times—patterns suggesting that strategic refinement of decision criteria, rather than (or in addition to) improvements in evidence accumulation, may also characterize response inhibition development. These dissociations suggest fundamentally different developmental mechanisms and provide the mechanistic precision that traditional behavioral approaches cannot offer. Rather than invoking vague notions of "poor inhibitory control," computational parameters specify whether children struggle with evidence quality (drift rate), response caution (boundaries), or prepotent response regulation (starting point)—distinctions with direct implications for targeted intervention.

The present study

Building on this theoretical and methodological foundation, the present study addressed two aims. The first aim (primary confirmatory) was to characterize the developmental trajectories of interference and response inhibition from childhood through adolescence. Based on multicomponent EF theory (Friedman & Miyake, 2017; Miyake et al., 2000; Nigg, 2000) and neurodevelopmental evidence (Luna et al., 2015), we predicted that both components would show nonlinear development (Hypothesis 1a) with response inhibition plateauing earlier than interference inhibition (Hypothesis 1b). The second aim (primarily exploratory) was to identify the computational mechanisms underlying developmental changes in each inhibitory component. Based on DDM research on inhibitory control (Ambrosi et al., 2020; Duc et al., 2025; Ratcliff et al., 2012), we hypothesized that interference and response inhibition would show partially distinct computational signatures, with some parameters exhibiting shared developmental patterns across both components and others showing task-specific trajectories (Hypothesis 2). Additionally, given that the factor structure of EF progressively differentiates from a unitary architecture in early childhood to separable components by adolescence (Lee et al., 2013; Shields & Yonelinas, 2025; Usai et al., 2014), we further explored whether the balance between shared and task-specific computational mechanisms shifts across development—hypothesizing that younger children would show more domain-general patterns, while older adolescents would show greater task-specific differentiation (Hypothesis 3).

To achieve these aims, we recruited participants aged 6–18 years from two independent sites (see Figure 1). By integrating flexible nonlinear modeling with computational decomposition, this approach allowed us to move beyond describing *when* developmental changes occur to explaining *how* they unfold—providing empirical traction on the unity-versus-diversity debate and informing component-specific intervention strategies.

Method

Participants

We recruited 2,148 children and adolescents (aged 6–18 years) from schools across two regions in China. Participants were recruited from two independent sites: a primary sample (n = 1,122; 45.54% female; M_{age} = 12.42 years, SD = 3.79 years, range: 6.0–18.7 years) with longitudinal follow-up and a validation sample (n = 1,026; 45.13% female; M_{age} = 12.44 years, SD = 3.77 years, range: 6.1–18.8 years) for cross-sectional replication. Both samples were predominantly Han Chinese (primary: 91.9%; validation: 90.8%). The two sites demonstrated comparable

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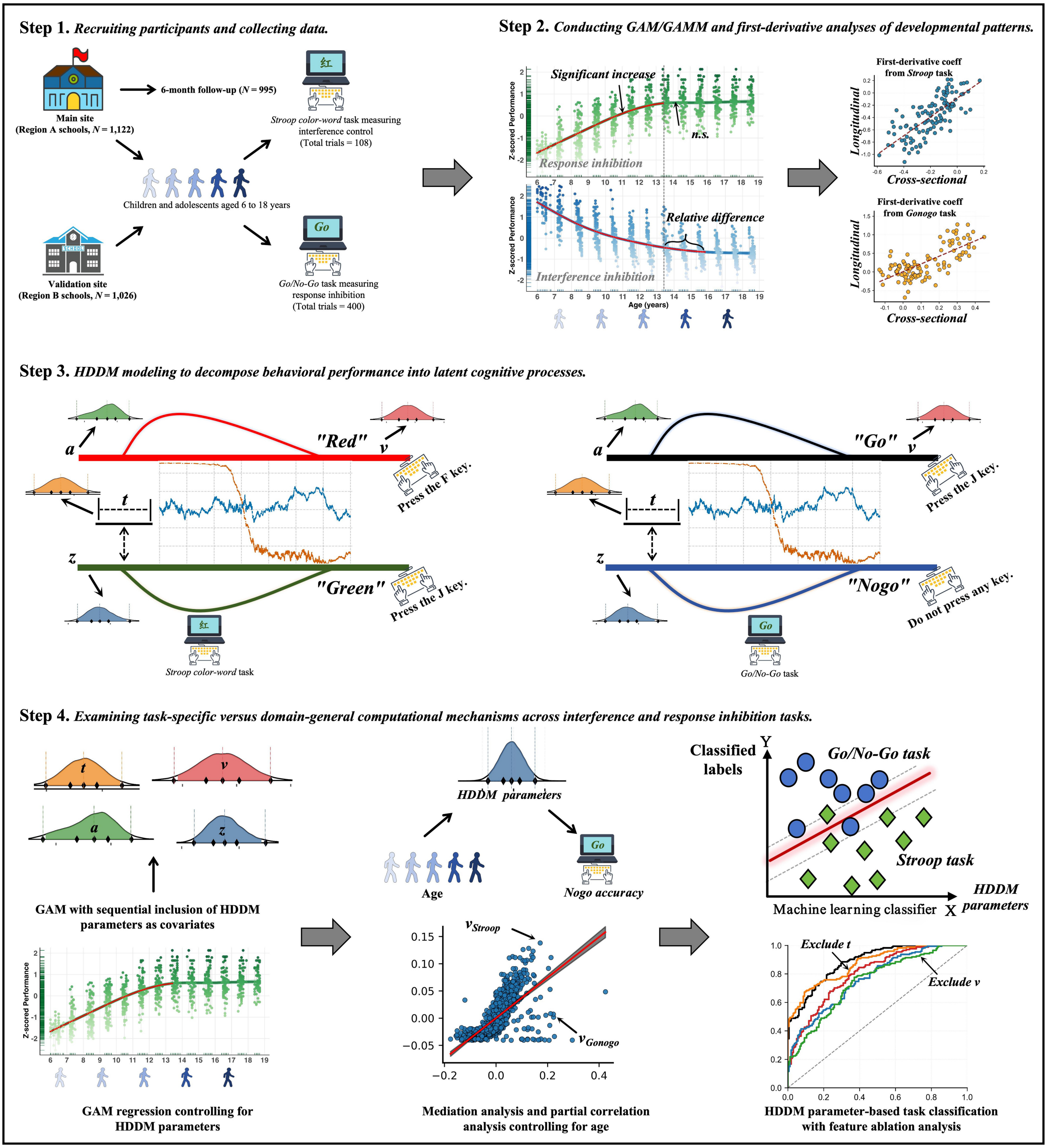

Figure 1 Overview of study design and analytical pipeline. This figure illustrates the four-stage analytical framework. Step 1. Participants were recruited from two independent sites—a primary sample ($n = 1{,}122$) with 6-month longitudinal follow-up ($n = 995$) and a validation sample ($n = 1{,}026$)—and completed the *Stroop color–word* task (interference inhibition) and *Go/No-Go* task (response inhibition), along with sociodemographic measures. Step 2. Cross-sectional patterns were characterized using generalized additive modeling, and longitudinal stability was assessed using generalized additive mixed modeling; first-derivative analyses identified periods of statistically significant developmental change. Step 3. Trial-level behavioral data were analyzed using the hierarchical drift diffusion model to estimate four core decision parameters: drift rate (v), decision boundary (a), starting point bias (z), and nondecision time (t). Step 4. Task-specific and shared computational mechanisms were examined through covariate-adjusted generalized additive modeling, bootstrap mediation analyses, age-controlled partial correlations, and support vector machine classification with feature ablation.

demographic composition across all age groups (Table S4), with no significant between-site differences in baseline distributions of inhibitory control measures (all $ps > .05$, Kolmogorov–Smirnov tests; Figure S1). Data were collected between September 2022 and July 2024 in public schools in Gansu Province, mainland China. The study protocol was approved by the Ethics Committee

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of Northwest Normal University (Protocol code: NWNU202001). To examine longitudinal developmental change, we conducted 6-month follow-up assessments of the primary sample. Of the original 1,122 participants, 995 (88.68%) completed follow-up testing. Attrition rates varied across age groups, from 6.25% among 6-year-olds to 14.47% among 18-year-olds, with no significant age-related differences ($\chi^2(12)=5.92$, $p=.920$). Attrition analyses revealed no selection bias: participants who completed follow-up did not differ from those lost to attrition on baseline *Stroop* interference ($t=0.83$, $p=.405$) or *Go/No-Go* accuracy ($t=1.02$, $p=.307$).

Inclusion criteria required participants to: (a) be native Mandarin Chinese speakers; (b) have normal or corrected-to-normal vision; (c) be free from parent-reported neurological disorders, psychiatric conditions, or cognitive developmental abnormalities; and (d) have no prior experience with similar cognitive tasks. We excluded participants who were taking medications known to affect cognitive function or who had been diagnosed with learning difficulties or attention-deficit/hyperactivity disorder. Parents or legal guardians provided written informed consent, and participants provided age-appropriate assent. All procedures followed the ethical principles of the Declaration of Helsinki. Participants received a small gift as compensation.

Materials and procedure

We used the *Stroop* and *Go/No-Go* tasks for our study, as they are among the most widely used measures of interference inhibition and response inhibition, respectively (Nigg, 2000), and tap theoretically distinct subcomponents of inhibitory control (Diamond, 2013). Both paradigms have clearly defined neural substrates (Laird et al., 2005; Simmonds et al., 2008), demonstrate robust developmental sensitivity across childhood and adolescence (Cragg & Nation, 2008; Davidson et al., 2006; Homack & Riccio, 2004), and were administered with identical stimulus materials across the 6–18-year age range—as the HDDM is sensitive to variations in stimulus properties, using age-adapted paradigms would introduce systematic confounds in extracted parameters, precluding valid cross-age comparisons (Wiecki et al., 2013).

Stroop task

We assessed interference inhibition using a computerized adaptation of the classical *Stroop* paradigm (Stroop, 1935). The stimuli consisted of two high-frequency Chinese color words ("红" [red], "绿" [green]) and neutral symbols ("####"), presented centrally in either red or green ink (see Figure S2). These color words were selected based on their early acquisition in Chinese language development—by first grade (typically age 6 years), Chinese children have mastered these characters and their meanings—supporting task validity across our age range, consistent with previous developmental studies in Chinese populations (Gui et al., 2020; Ji et al., 2011; Su et al., 2023; Wu et al., 2011). Participants identified the ink color while ignoring the word meaning, pressing "F" for red (left index finger) or "J" for green (right index finger). The task included three conditions: (*a*) congruent, where the word meaning matched the ink color (e.g., "红" in red ink); (b) incongruent, where the word meaning conflicted with the ink color (e.g., "绿" in red ink); and (c) neutral, where symbols without semantic content were present (e.g., "####" in red ink). The *Stroop* interference effect, calculated as the reaction time difference between incongruent and congruent conditions, served as the primary measure of interference inhibition (Rey-Mermet et al., 2018). Each trial followed a fixed sequence: fixation cross (500 ms) → blank screen (1,000 ms) → target stimulus (maximum 1,500 ms) → intertrial interval (500 ms). Following 18 practice trials with feedback, participants who met the 85% accuracy criterion—ensuring adequate task comprehension and reliable performance across our full age range—proceeded to three blocks of 36 trials each (12 per condition), totaling 108 experimental trials. Conditions were pseudorandomized with no more than three consecutive trials of the same type. Self-paced rest breaks between blocks limited the total task duration to ~15 min. The *Stroop* interference effect demonstrated good internal consistency in prior research (Cronbach's $\alpha=.84$; Zhu et al., 2025).

Go/No-Go task

Response inhibition was measured using a letter-based *Go/No-Go* paradigm (Gomez et al., 2007). Simple letter stimuli ("X" and "Y") minimized semantic processing demands, ensuring accessibility across all ages (Lahat et al., 2010; Lewis et al., 2017; Xie et al., 2022). Participants responded rapidly to *Go* stimuli (target letters) while withholding responses to *No-Go* stimuli (nontargets; see Figure S2). The brief stimulus duration (600 ms) and response window (400 ms) enhanced prepotent response tendencies, thereby increasing sensitivity to inhibitory control differences. The trial structure consisted of: fixation cross (1,000 ms) → stimulus (600 ms) → blank response window (400 ms) → intertrial interval (500 ms). To control for stimulus-specific confounds, we employed a counterbalanced target design across four blocks of 100 trials each (75% *Go*, 25% *No-Go*). Target assignments alternated between blocks: X served as the *Go* stimulus in blocks 1 and 3, and Y in blocks 2 and 4. Block order was counterbalanced across participants using a Latin-square design. Following 20 practice trials with feedback, participants who met the same 85% accuracy criterion proceeded to 400 experimental trials. The primary outcome measures included *Go* trial reaction time and accuracy; *No-Go* trial accuracy; and d', a signal detection measure of discriminability calculated as $d'=z(\text{hit rate})-z(\text{false alarm rate})$ (Young et al., 2018). The task demonstrated high internal consistency (Cronbach's $\alpha=.88$) and required ~15 min to complete in prior research (Zhu et al., 2025).

Testing environment

Testing took place in school computer laboratories using standardized 15.6-inch LED monitors (1,600 × 900 resolution, 60-Hz refresh rate) positioned 60 cm from participants. *E*-Prime 2.0 (Psychology Software Tools, Pittsburgh, PA, USA) controlled stimulus presentation and recorded responses with millisecond-level precision. Trained graduate students in psychology administered all sessions following standardized protocols. Experimenters confirmed participant understanding before formal testing.

Data analytic strategy

Modeling nonlinear patterns and identifying critical developmental periods

Participants whose reaction times fell more than three standard deviations from their age group means were excluded (Ambrosi et al., 2020). We employed GAMs for cross-sectional analyses and

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generalized additive mixed models (GAMMs) for longitudinal analyses (Hao et al., 2025; Tervo-Clemmens et al., 2023). These approaches use data-driven smooth functions to capture nonlinear developmental patterns without imposing predetermined functional forms, thereby enabling detection of developmental accelerations, decelerations, and plateaus that polynomial regression might miss (Hastie, 2017). GAMMs additionally incorporate random effects to account for within-participant dependencies (see Supplementary Methods for details). To identify periods of significant developmental change, we calculated first derivatives of the fitted curves at 0.1-year intervals and generated simultaneous 95% confidence intervals through 10,000 simulations (Simpson, 2018). Significant developmental periods were defined as age ranges where these confidence intervals excluded zero, indicating reliable rates of change in inhibitory control (Tervo-Clemmens et al., 2023).

HDDM analysis

We employed the HDDM (version 1.0.1) implemented in Python to conduct computational modeling of trial-level behavioral data from the *Stroop* and *Go/No-Go* tasks (Pan et al., 2025). We implemented two hierarchical models. The baseline model included four core parameters—decision boundary (a), drift rate (v), nondecision time (t), and starting point bias (z) (Ratcliff & McKoon, 2008)—with uniform distributions across conditions and age groups. The age-effect model incorporated age as a continuous predictor, allowing all parameters to vary with age to capture developmental patterns (Ratcliff et al., 2012). We used default prior distributions from the HDDM package, derived from meta-analytic findings, which balance conservatism against flexibility (Pan et al., 2025; Zhang et al., 2025). Parameter estimation used Markov Chain Monte Carlo (MCMC) sampling within a hierarchical Bayesian framework (Wiecki et al., 2013). Each model was run with four independent chains of 15,000 iterations, with the first 5,000 discarded as burn-in. Convergence was verified using the Gelman–Rubin statistic (<̂ 1.02; Gelman et al., 2014) and effective sample sizes (> 400; Pan et al., 2025). Model validity was evaluated through posterior predictive checks comparing 500 simulated datasets with observed data (Peters & D'Esposito, 2020). For statistical inference, we used Bayes factors (BF_{10}) to quantify evidence strength and 95% highest density intervals (HDIs) with regions of practical equivalence (ROPEs) to characterize parameter uncertainty (Kruschke & Liddell, 2018) (see Supplementary Methods).

Examining computational mechanisms underlying inhibitory control

To examine similarities and differences in the computational mechanisms underlying the two forms of inhibitory control, we analyzed parameter specificity and commonality. For specificity analysis, we first employed bootstrap mediation analysis (3,000 samples) to examine whether HDDM parameters mediated the relationship between age and behavioral performance (Liu et al., 2022). We then constructed nested GAM models, sequentially controlling for HDDM parameters, and compared Akaike information criterion values to quantify each parameter's explanatory contribution to developmental patterns (Burnham & Anderson, 2002). For commonality analysis, we calculated partial correlations between corresponding parameters across tasks, controlling for age. We also constructed a support vector machine (SVM) classifier (Wiecki et al., 2015) using the four core HDDM parameters to discriminate task type (*Stroop* vs. *Go/No-Go*), and employed feature ablation analysis to quantify each parameter's contribution to task discrimination (Biazoli Jr et al., 2024) (see Supplementary Methods for details).

Results

Measurement validation across development

Before examining developmental patterns, we validated task appropriateness and measurement equivalence across ages 6–18 years. Both tasks demonstrated robust psychometric properties across two independent samples, with performance metrics remaining within appropriate ranges throughout development. Split-half reliability was acceptable across all age groups (*Stroop*: $r = .75$–$.90$; *Go/No-Go*: $r = .85$–$.95$; Tables S5 and S6), and test–retest reliability from the 6-month follow-up was robust across all four developmental periods ($r = .648$–$.964$; Figures S3 and S4). To further establish measurement equivalence, we conducted multigroup confirmatory factor analysis across the four developmental periods (Table S7). Configural, metric, and scalar invariance were all supported ($\Delta CFI < .01$, $\Delta RMSEA < .015$; Chen, 2007). We also verified that the HDDM framework maintained adequate model fit across all age groups (Table S8). Together, these results confirm that both tasks and computational models provide equivalent measurements across the full age range.

Testing Hypothesis 1: Developmental patterns of inhibitory control from ages 6–18 years

Nonlinear patterns and critical developmental periods with cross-site validation

To test Hypotheses 1a and 1b, we applied GAMs with penalized splines to fit cross-sectional data from children and adolescents aged 6–18 years. All GAM models showed nonlinear patterns (effective degrees of freedom [EDF] range: 2.99–8.03, all $ps < .001$), indicating that nonlinear models provided better fit than linear alternatives and supporting Hypothesis 1a (Table S9). To identify periods of significant change in inhibitory control abilities, we calculated first derivatives for each GAM model at 0.1-year intervals. Through simulation (10,000 iterations), we constructed simultaneous confidence intervals for first derivatives to account for multiple comparisons, evaluating statistical significance at each age point ($p < .05$, two-sided).

Analysis of developmental patterns revealed that inhibitory control abilities showed continuous but uneven age-related changes between ages 6 and 18 years, characterized by four distinct periods (Figure 2A, B; Table S10): (1) Early childhood rapid change (6–8 years): This period exhibited the steepest improvements across all measures, with both interference and response inhibition showing marked age-related gains; (2) Middle childhood sustained growth (9–12 years): During this period, developmental rates decelerated to ~50% of the early phase rate, though steady progress continued; (3) Midadolescent deceleration (13–15 years): Further slowing characterized this

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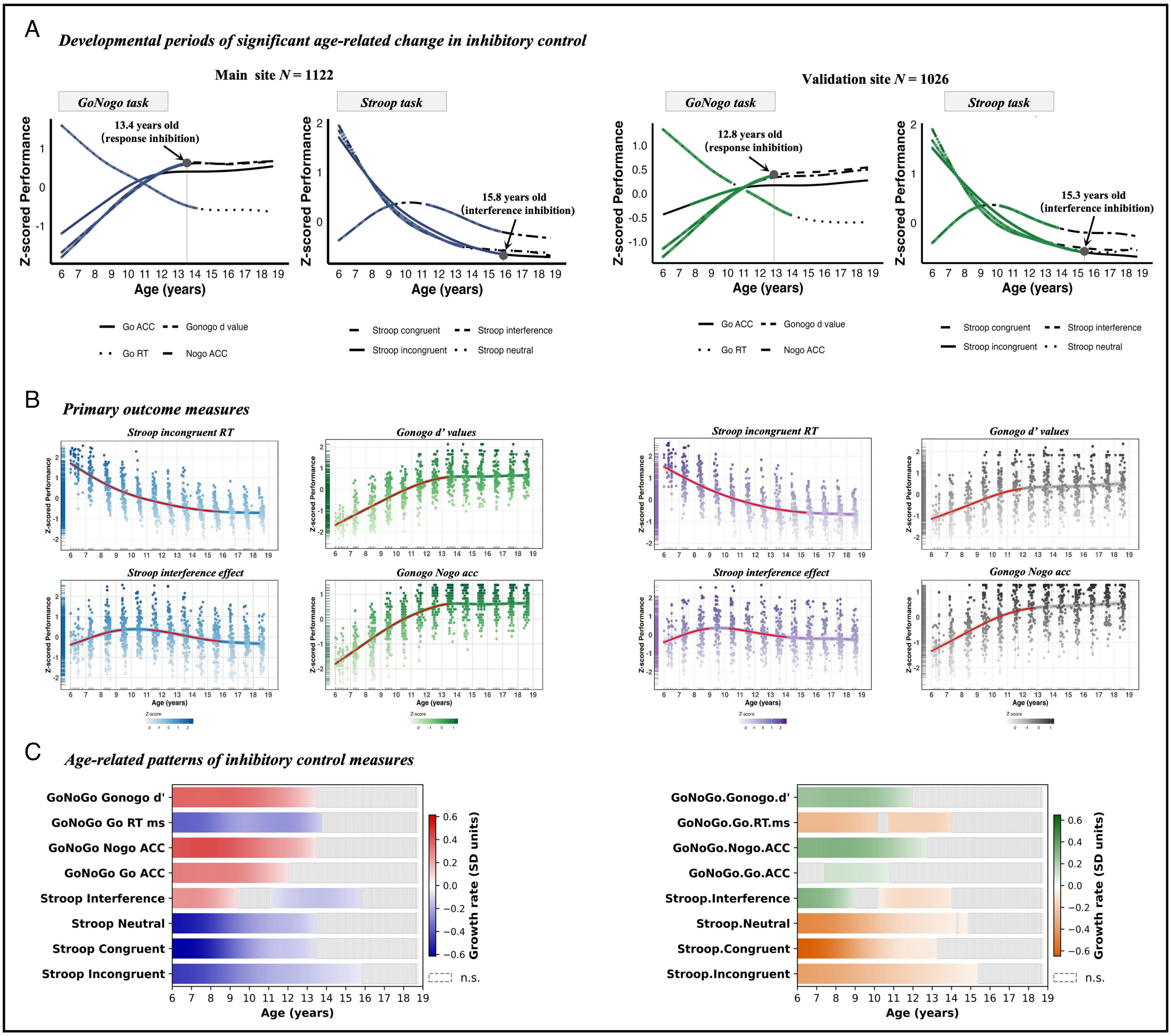

Figure 2 Developmental patterns of inhibitory control across primary and validation samples. (A) Generalized additive model–fitted curves for interference inhibition (*Stroop* task) and response inhibition (*Go/No-Go* task) across ages 6–18 years. The left column presents the primary sample; the right column presents the validation sample. Colored segments indicate age periods of statistically significant change ($p < .05$, simultaneous confidence intervals corrected for multiple comparisons); black segments denote nonsignificant periods. (B) Performance trajectories for individual indicators with 95% confidence intervals. Red markers indicate statistically significant age-related change; nonred markers indicate nonsignificant change. (C) Heat maps of first-derivative values (instantaneous rates of change) for each inhibitory control measure across age. Warm colors indicate significant improvement; cool colors indicate significant decline; white indicates nonsignificant change. RT = reaction time; Acc = accuracy.

transitional period, with all indicators showing modest but significant improvements; and (4) Late adolescent stabilization (16–18 years): Developmental changes during this period ceased to reach statistical significance, indicating that inhibitory control abilities plateaued by age 16 years within our observation window (Figure 2C).

Consistent with Hypothesis 1b, interference inhibition and response inhibition demonstrated distinct developmental timelines. The significant developmental period for the *Stroop* incongruent condition extended to 15.8 years, representing 9.8 years of significant change (77.17% of the 12.7-year observation period). Response inhibition reached its plateau earlier, with *No-Go* accuracy and *d′* showing significant changes only until 13.4 years, representing 7.4 years of significant change (58.27% of the observation period). Bootstrap analysis confirmed that interference inhibition (*Stroop* incongruent) and response inhibition (*No-Go* accuracy) differed significantly in their plateau timing by 2.4 years (95% CI [1.84, 2.96], p_{FDR} (false discovery rate corrected) < .001; Figure S5A). Notably, the *Stroop* interference effect followed an inverted U-shaped developmental pattern, with interference magnitude initially increasing before declining (Figure 2A, B).

The validation sample replicated both the four-phase nonlinear pattern and differential timing between interference and response inhibition (Figure 2A, B; Table S9). Interference inhibition again showed significantly more protracted development

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than response inhibition, with a mean difference of 2.5 years (95% CI [1.94, 3.06], p_{FDR} < .001; Figure S5B). Thus, both Hypotheses 1a and 1b were supported and replicated across independent samples.

Longitudinal validation with GAMM

To validate the developmental patterns through within-subject changes, we conducted 6-month follow-up assessments with the primary sample (Figure 3A). The four-phase nonlinear developmental pattern was confirmed through GAMM (Tables S9 and S10), and cross-sectional GAM and longitudinal GAMM derivatives showed strong concordance (r = .627–.749, p_{FDR} < .001; Figure 3B), supporting the validity of the cross-sectional patterns. Longitudinal GAMM also confirmed differential developmental timing: interference inhibition showed significant changes until ~16 years, while response inhibition plateaued around 15 years (mean difference = 1.1 years, 95% CI [0.54, 1.66], p_{FDR} < .001; Figure S5C). In summary, longitudinal validation provided convergent support for both Hypotheses 1a and 1b.

Testing Hypothesis 2: Distinct computational mechanisms for interference and response inhibition

To test the hypothesis regarding partially distinct computational signatures (Hypothesis 2), we applied HDDM to trial-level behavioral data across both tasks. Given adequate model fit across age groups (see section "Measurement Validation Across Development"), we proceeded with parameter estimation (convergence diagnostics in Figures S6 and S7; see Supplementary Results for details). Bayesian analysis of variance revealed significant between-task differences in computational parameters. The drift rate was substantially higher in the *Stroop* task (95% HDI [0.14, 0.37]; $BF_{10} = \infty$; $PP|D = .002$; 0.4% in ROPE; Figure 4A), suggesting more efficient evidence accumulation during interference resolution compared with response inhibition. Starting point bias showed the opposite pattern, with lower values in the *Stroop* task relative to the *Go/No-Go* task (95% HDI [−0.46, −0.41]; $BF_{10} = \infty$;

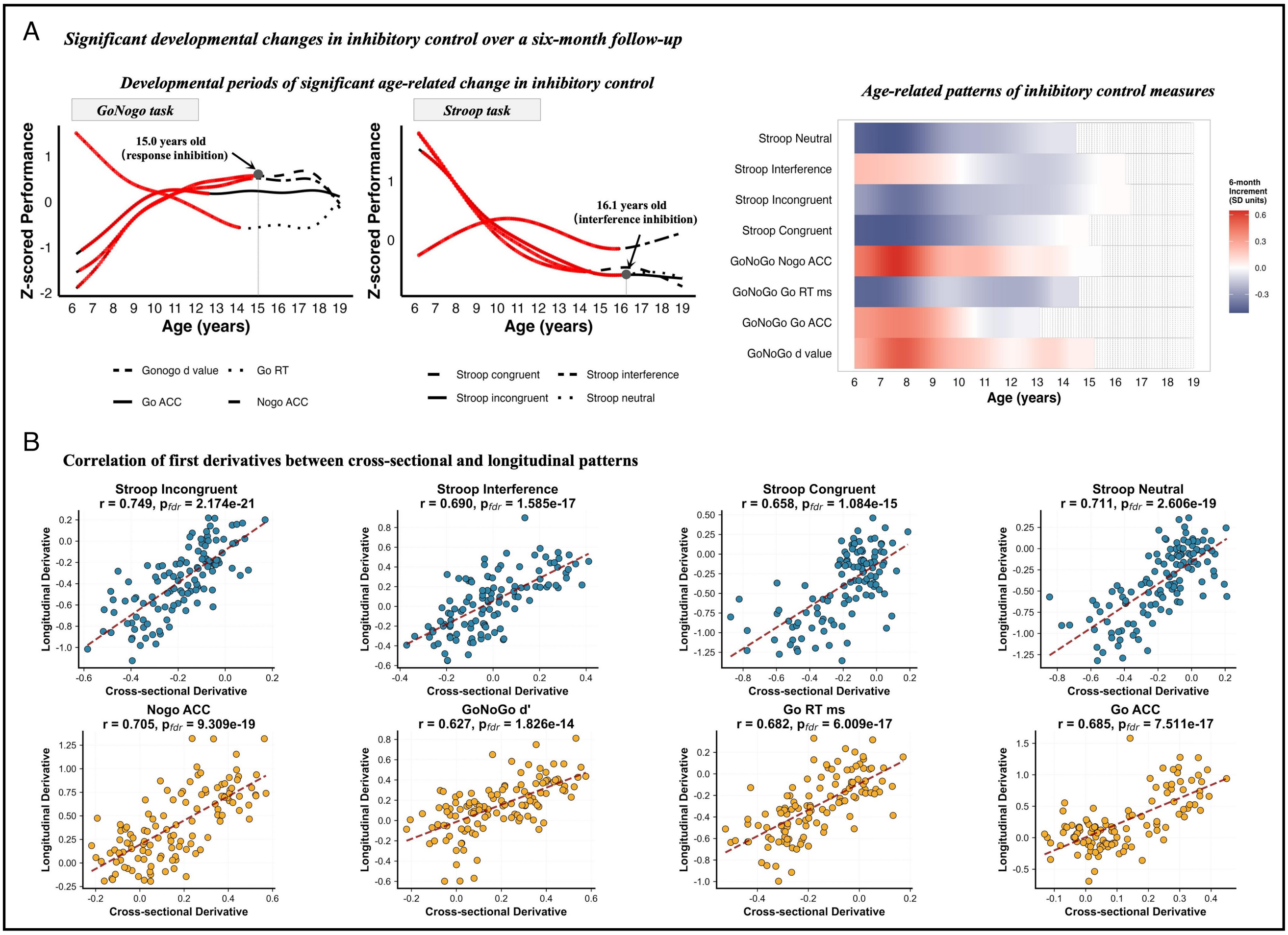

Figure 3 Longitudinal validation of developmental patterns through 6-month follow-up assessment. (A) Generalized additive mixed model results from the longitudinal subsample (n = 995). The left panel displays fitted developmental curves for interference inhibition (*Stroop* task) and response inhibition (*Go/No-Go* task), with colored segments indicating statistically significant change (p < .05) and black segments indicating nonsignificant periods. The right panel presents heat maps of first-derivative values (instantaneous rates of change); warm colors indicate significant improvement, cool colors indicate significant decline, and white indicates nonsignificant change. (B) Correspondence between cross-sectional (generalized additive modeling) and longitudinal (generalized additive mixed modeling) first-derivative estimates across age. Each point represents the estimated rate of change at a given age; blue denotes interference inhibition measures, while orange denotes response inhibition measures. RT = reaction time; Acc = accuracy.

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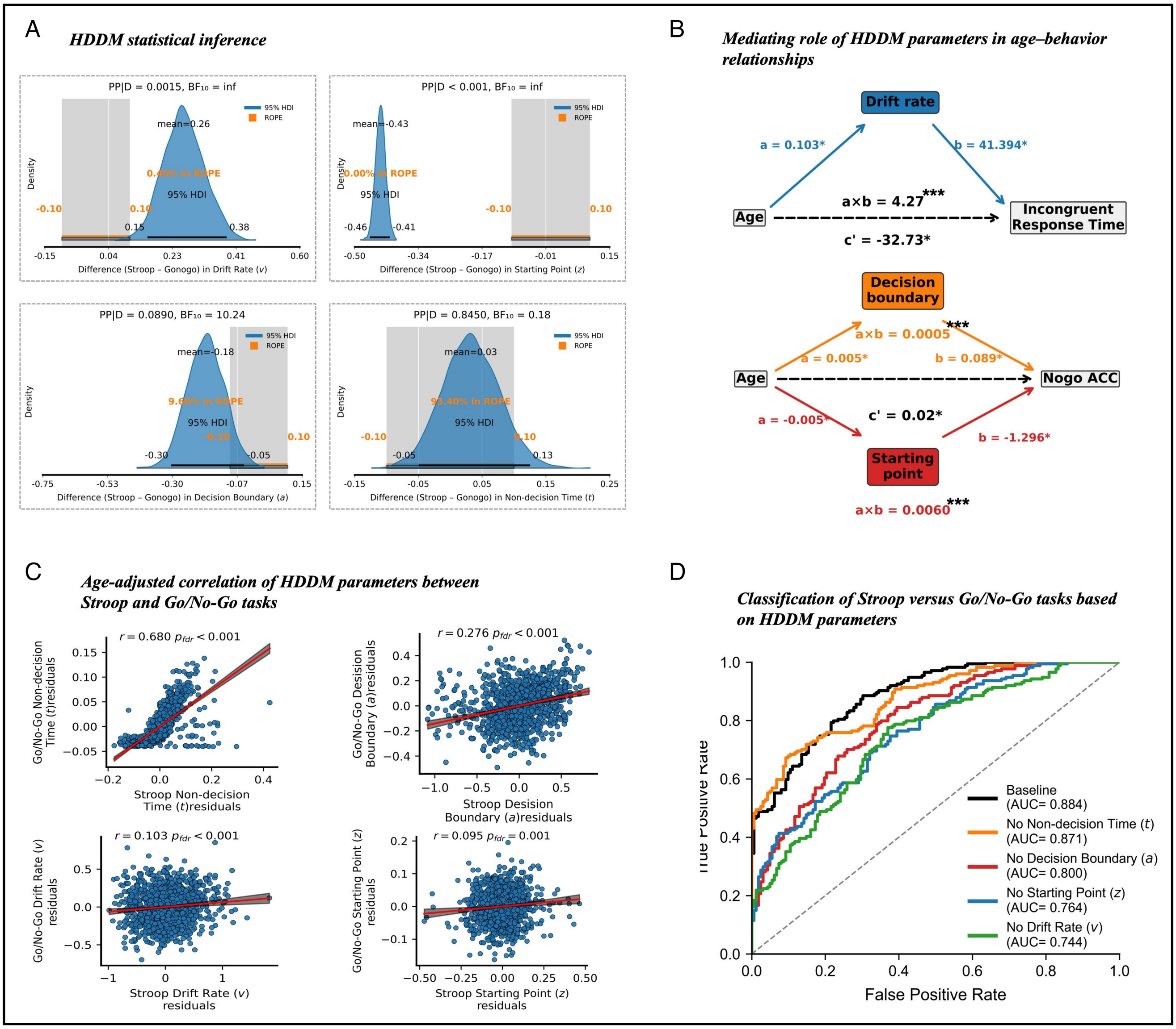

Figure 4 Dissociating task-specific and shared computational mechanisms underlying inhibitory control. (A) Bayesian contrasts of hierarchical drift diffusion model (HDDM) parameters between *Stroop* and *Go/No-Go* tasks. Posterior distributions display between-task differences for each parameter; values outside the region of practical equivalence indicate meaningful differences. (B) Bootstrap mediation analyses examining HDDM parameters as mediators of age-related performance improvements. Only statistically significant mediation pathways are displayed. (C) Age-adjusted partial correlations between corresponding HDDM parameters across tasks. (D) Support vector machine classification performance distinguishing *Stroop* from *Go/No-Go* task data based on HDDM parameters, with feature importance quantified through systematic ablation analysis. a = decision boundary; PP|D = posterior probability given data; AUC = area under the receiver operating characteristic curve; HDI = highest density interval; ROPE = region of practical equivalence; t = nondecision time; v = drift rate; z = starting point bias. *$p < .05$. **$p < .01$. ***$p < .001$.

PP|D < .001; 0.0% in ROPE). The decision boundary was also lower in the *Stroop* task compared with the *Go/No-Go* task (95% HDI [−0.32, −0.07]; $BF_{10} = 10.2$; *PP*|D = .089; 10.2% in ROPE). Nondecision time showed no meaningful difference between tasks (95% HDI [−0.06, 0.12]; $BF_{10} = 0.18$; 93.4% in ROPE).

Mediation analysis revealed a significant negative association between age and incongruent reaction time ($\beta = -28.46$, $p < .001$; Figure 4B), indicating improved conflict-resolution ability across development. Drift rate (v) emerged as a significant mediator (indirect effect 95% CI [1.97, 6.56], $p < .001$), suggesting that age-related improvements in conflict processing speed are associated with enhanced efficiency in evidence accumulation. While decision boundary (a) increased significantly with age ($\beta = .012$, $p < .001$), reflecting developmental increases in response caution, its direct effect on reaction time was not significant ($\beta = -25.46$, $p = .221$), resulting in a nonsignificant indirect effect (95% CI [−0.88, 0.23], $p = .267$). The indirect effects of nondecision time (t) and starting point bias (z) were also nonsignificant. The results for the *Go/No-Go* task revealed a contrasting pattern. Age positively predicted *No-Go* accuracy ($\beta = .018$, $p < .001$), reflecting developmental improvements in response inhibition. Unlike the *Stroop* task, starting point bias (z) served as the primary mediator, accounting for 34.0% of the age effect. Specifically, age was associated with reduced starting point bias ($\beta = -.005$, $p < .001$), indicating a developmental decrease in prepotent “go” response tendencies. This bias reduction was significantly associated with

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enhanced *No-Go* accuracy through attenuating impulsive responses ($\beta = -1.296$, $p < .001$), yielding a significant positive indirect effect (0.006, 95% CI [0.0053, 0.0068], $p < .001$). Decision boundary (*a*) demonstrated a significant but smaller mediating role, contributing 2.64% to the total effect. Age-related increases in decision criteria ($\beta = .005$, $p < .001$) were associated with improved task performance through more conservative responses ($\beta = .089$, $p < .001$), producing an indirect effect of 0.0005 (95% CI [0.0003, 0.0007], $p < .001$). Neither drift rate (*v*) nor nondecision time (*t*) showed significant mediating roles in the *Go/No-Go* task, supporting the hypothesis of task-specific computational mechanisms.

GAM analyses with systematic parameter control converged with mediation findings (Table S11): drift rate accounted for the largest share of age-related variance in *Stroop* performance ($\Delta R^2 = -.128$, 24.5% of the original age effect), while starting point bias contributed most to *Go/No-Go* performance ($\Delta R^2 = -.124$, 19.1%). Decision boundary made moderate contributions in both tasks (*Stroop*: 12.6%; *Go/No-Go*: 14.2%), and nondecision time showed comparable contributions across tasks (*Stroop*: 10.3%; *Go/No-Go*: 10.8%). Notably, starting point bias had minimal effects on the *Stroop* task (2.3%), while drift rate made limited contributions to the *Go/No-Go* task (5.1%). After controlling for all HDDM parameters simultaneously, the models accounted for ~40.2% and 35.5% of the age-related variance in *Stroop* and *Go/No-Go* performance, respectively, and nonlinear age effects in both tasks were no longer significant ($p > .05$).

Partial correlation analyses further explored the developmental associations of each parameter between tasks. Nondecision time (*t*) showed the strongest cross-task association ($r = .680$, $p_{FDR} < .001$; Figure 4C), confirming shared improvements in processing speed across both inhibitory components. Decision boundary (*a*) exhibited a moderate but significant cross-task association ($r = .276$, $p_{FDR} < .001$). In contrast, drift rate (*v*) and starting point bias (*z*) showed pronounced task specificity. Notably, the cross-task association for drift rate decreased markedly when age was controlled (from $r = .364$, $p_{FDR} < .001$ in zero-order correlation to $r = .103$, $p_{FDR} = .001$ in partial correlation). Similarly, starting point bias demonstrated minimal cross-task association ($r = .095$, $p_{FDR} = .001$), further confirming the task-specific nature of response preparation strategies.

To complement these correlational findings and quantify the degree of task specificity for each parameter, we employed SVM classifiers for feature importance analysis. The baseline model (including four parameters: *a*, *v*, *t*, and *z*) effectively distinguished between the *Stroop* and *Go/No-Go* tasks on the test set through 5-fold × 3-fold inner nested cross-validation (area under the receiver operating characteristic curve [AUC] = .884; Figure 4D). Systematic parameter removal revealed that drift rate (*v*) and starting point bias (*z*) showed strong task discriminative ability: removing drift rate led to a significant decrease in classification performance (AUC = .744, ΔAUC = −.140, $p_{FDR} = .004$), and removal of starting point bias similarly resulted in a significant decrease (AUC = .764, ΔAUC = −.120, $p_{FDR} = .004$). While ablation of the decision boundary parameter resulted in a statistically significant but modest decrease (AUC = .800, ΔAUC = −.084, $p_{FDR} = .008$), ablation of nondecision time had virtually no impact (AUC = .871, ΔAUC = −.013, $p_{FDR} = .349$). These findings were largely replicated in the independent validation sample (Figure S8).

Testing Hypothesis 3: Progressive differentiation of computational mechanisms across development

To test Hypothesis 3—that computational dissociations between interference and response inhibition emerge progressively across development—we divided participants into four developmental periods based on transition points identified in our GAM derivative analyses and established developmental frameworks (e.g., Best & Miller, 2010; Luna et al., 2015): early childhood (6–8 years, $n = 270$), middle childhood (9–12 years, $n = 344$), mid-adolescence (13–15 years, $n = 261$), and late adolescence (16–18 years, $n = 247$).

Bayesian analysis of variance revealed that between-task parameter differences emerged and strengthened progressively with age (Table S12; Figure S9). For drift rate, the pattern of higher values in the *Stroop* task relative to the *Go/No-Go* task was not reliably detected in early childhood ($BF_{10} = 1.8$), first emerged in middle childhood ($BF_{10} = 8.4$), and became well established by adolescence ($BF_{10} = 156$–∞). Starting point bias differences were already evident in early childhood ($BF_{10} = 42$) but continued to increase across subsequent periods. Decision boundary differences emerged only in adolescence ($BF_{10} = 5.6$–38). In contrast, nondecision time showed no between-task differences at any developmental period (all $BF_{10} < 0.35$). Mediation analyses confirmed that task-specific computational patterns remained consistent across all periods (Figure S10)—drift rate (*v*) was the sole significant mediator for *Stroop* performance, while starting point bias (*z*) served as the primary mediator for *Go/No-Go* performance, with effect sizes increasing progressively. Notably, decision boundary (*a*) also contributed to *Go/No-Go* performance during childhood ($p < .01$) before diminishing to nonsignificance by late adolescence, suggesting that younger children may rely on elevated decision thresholds as a compensatory strategy. Cross-task correlations provided additional evidence for progressive differentiation (Figure S11): both drift rate and starting point bias correlations declined from significant associations in early childhood (drift rate: $r = .286$; starting point bias: $r = .243$) to nonsignificance by adolescence (drift rate: $r = .052$–.078; starting point bias: $r = .042$), whereas nondecision time maintained robust associations throughout ($r = .583$–.714, all $ps < .001$). Converging with these findings, SVM classification accuracy for distinguishing between tasks improved substantially from early childhood to late adolescence (AUC: .712 to.906), and feature ablation revealed a developmental shift in discriminating features—decision boundary contributed most in early childhood (ΔAUC = −.091), whereas drift rate and starting point bias became dominant by late adolescence (ΔAUC = −.128 and −.124, respectively). These findings support Hypothesis 3, revealing a developmental shift from partial reliance on domain-general strategies in early childhood to more differentiated, task-specific computational processes by late adolescence.

Discussion

This study examined the developmental patterns and computational mechanisms of inhibitory control from ages 6–18 years using cross-sectional (combined $N = 2{,}148$), validation, and longitudinal designs. This represents the first study to provide computational evidence for both unity and diversity in inhibitory control

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development across childhood and adolescence through a decision-processing lens. We identified three key findings. First, both interference and response inhibition demonstrated nonlinear four-phase patterns—rapid improvement (6–8 years), sustained growth during preadolescence (9–12 years), deceleration in midadolescence (13–15 years), and stabilization in late adolescence (16–18 years). Second, these components exhibited distinct developmental trajectories, with response inhibition plateauing at 13.4 years, while interference inhibition continued improving until 15.8 years—a 2.4-year difference replicated across independent samples. Third, computational modeling revealed distinct underlying mechanisms: age-related changes in interference inhibition were primarily associated with enhanced drift rate (v), reflecting improved conflict resolution efficiency, whereas age-related changes in response inhibition were linked to systematic adjustments in starting point bias (z), indicating strategic response optimization. Both components showed shared improvements in nondecision time (t), suggesting common foundations in processing speed. Importantly, age-stratified analyses revealed that this computational dissociation emerges progressively across development—from partial reliance on domain-general threshold-based control in early childhood to more differentiated, task-specific computational processes by late adolescence.

Nonlinear development with distinct trajectories

Consistent with Hypothesis 1a, GAM analyses revealed that both interference and response inhibition demonstrated nonlinear developmental patterns from ages 6–18 years, characterized by four distinct phases. These findings align with previous investigations documenting nonlinear trajectories in EF development across childhood and adolescence (Best & Miller, 2010; Ferguson et al., 2021; Luna et al., 2004), with recent large-scale cross-site and multinational analyses providing further corroboration (Folker et al., 2025; Tervo-Clemmens et al., 2023). As Diamond (2013) observed, improvements in childhood inhibitory ability correspond closely with the maturation of prefrontal cortex structure and function—a process that undergoes substantial nonlinear transitions during adolescence (Fiske & Holmboe, 2019; Luna et al., 2015; Rubia, 2013). By separately examining response inhibition and interference inhibition, we found that both displayed nonlinear developmental patterns—a convergence that may reflect their overlapping yet partially distinct neural substrates. Both functions depend on prefrontal cortex activity, with response inhibition primarily engaging the right inferior frontal gyrus (Aron et al., 2014) and interference inhibition recruiting the anterior cingulate cortex and dorsolateral prefrontal regions (Laird et al., 2005). These regions undergo protracted structural and functional maturation during childhood and adolescence, including gray matter volume reductions and progressive white matter connectivity enhancement (Giedd & Rapoport, 2010; Gilmore et al., 2018).

These four developmental phases align with prior behavioral and neuroimaging research documenting distinct characteristics of inhibitory control at different ages. Early childhood (6–8 years) represents a period of rapid but variable inhibitory capacity: children demonstrate emerging but inconsistent performance, remaining susceptible to attentional lapses and environmental interference (Constantinidis & Luna, 2019; Davidson et al., 2006). Inhibitory control during this phase is fragile and context-dependent, heavily influenced by task demands and motivational factors (Garon et al., 2008). Middle childhood (9–12 years) is characterized by consolidation and increasing efficiency. Response inhibition approaches adult-like accuracy, reflecting maturation of motor inhibition circuits (Bunge et al., 2002; Huizinga et al., 2006). However, interference inhibition continues to develop as children acquire the capacity to maintain goal-relevant information while suppressing competing responses (Best & Miller, 2010). Adolescence (13–18 years) involves continued refinement rather than dramatic gains, particularly for complex cognitive control processes requiring integration across multiple brain systems (Constantinidis & Luna, 2019; Luna et al., 2015). Midadolescence (13–15 years) shows decelerating but still significant improvements, while late adolescence (16–18 years) approaches stabilization (Tervo-Clemmens et al., 2023). Throughout this period, increasing functional specialization and more efficient prefrontal recruitment support ongoing development, with interference inhibition showing the most protracted improvements, consistent with continued prefrontal maturation (Ordaz et al., 2013; Velanova et al., 2008).

Beyond these general nonlinear patterns, we observed a distinctive inverted U-shaped developmental trajectory for the *Stroop* interference effect, consistent with previous findings (Forte et al., 2024; Leon-Carrion et al., 2004). During the rising flank (~6–10 years), rapid reading automatization (Castles et al., 2018) and semantic network expansion (Nation, 2017) increase children's sensitivity to semantic–color conflicts. However, the still-developing nature of the prefrontal executive control system (Diamond, 2013) limits their ability to inhibit prepotent responses, creating a critical gap between conflict detection and resolution. This developmental asynchrony reflects differential maturation rates across brain regions: both the anterior cingulate cortex (responsible for conflict monitoring) and the dorsolateral prefrontal cortex (crucial for conflict resolution) continue to mature through late adolescence (Luna et al., 2015; Ordaz et al., 2013).

A critical distinction emerged in the timing of developmental stabilization: interference inhibition followed a more protracted course, extending to ∼15.8 years of age, whereas response inhibition stabilized around 13.4 years of age, consistent with Hypothesis 1b. This finding aligns with developmental sequence frameworks wherein basic inhibitory functions (e.g., motor inhibition) mature before more complex forms (Klenberg et al., 2001). Previous research has documented similar differential patterns (Martin-Rhee & Bialystok, 2008; Tiego et al., 2018). For instance, *Go/No-Go* task performance reaches stability around age 12 years (Brocki & Bohlin, 2004), whereas interference inhibition in *Stroop* tasks continues improving through late adolescence (Leon-Carrion et al., 2004; Yeung et al., 2020). However, these earlier studies predominantly employed between-group comparisons, limiting continuous age analyses. The present investigation assessed both components within a large-scale sample, providing more precise estimates through continuous modeling. This asynchronous pattern likely reflects differences in neurobiological foundations. Response inhibition primarily engages the ventral prefrontal–basal ganglia circuit, which shows relatively rapid maturation during childhood (Luna et al., 2015; Rubia, 2013). In contrast, interference inhibition requires coordinated integration across frontoparietal control networks that continue maturing into late adolescence (Crone & Steinbeis, 2017).

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Electrophysiological evidence supports this distinction, revealing differential neural maturation patterns between these inhibitory components (Brydges et al., 2013). Brydges and colleagues proposed that children's diffuse frontal activation reflects ongoing neural network organization, requiring broader recruitment to accomplish task demands. This reorganization from diffuse to focused activation (Constantinidis & Luna, 2019; Durston et al., 2006) may underlie the protracted trajectory of interference inhibition.

The pattern of protracted development observed here shows both convergence and divergence with findings from the temperament literature. While the inhibitory control facet of effortful control similarly shows continued maturation from preschool through adolescence (Atherton et al., 2020; Geeraerts et al., 2021), questionnaire-based studies typically report linear increases with no clear developmental plateau. This discrepancy likely reflects methodological differences: performance-based tasks index cognitive efficiency under controlled conditions, whereas temperament questionnaires capture emotional regulation and everyday behavior (Duckworth & Kern, 2011; Toplak et al., 2013). Indeed, correlations between task-based and questionnaire measures of inhibitory control are often modest (Duckworth & Kern, 2011). Nevertheless, both traditions converge on the conclusion that inhibitory capacity undergoes prolonged development beyond childhood, underscoring adolescence as a critical period for inhibitory control maturation.

Computational dissociation between interference and response inhibition

Existing research on inhibitory control development has remained primarily descriptive, offering limited insight into the underlying computational mechanisms that drive age-related changes (Hedge et al., 2018; Löffler et al., 2024; Luna et al., 2015). We addressed this gap by applying DDMs to examine age-related changes through a computational decision-processing lens. HDDM analysis of trial-level *Stroop* task data revealed that age-related improvements in interference inhibition were primarily associated with systematic increases in drift rate (v), supporting the hypothesis of task-specific computational mechanisms (Hypothesis 2). Drift rate reflects the efficiency of evidence accumulation—the speed and quality of stimulus processing (Hedge et al., 2018). In the *Stroop* task, successful performance requires selecting relevant information from competing sources and resolving cognitive conflict. Young children's prolonged reaction times appear closely related to differences in evidence-accumulation processes (Ambrosi et al., 2019). Supporting this interpretation, Lewis et al. (2017) found that younger children are more prone to attentional lapses, which could impair sampling efficiency and alter accumulation patterns. Our findings suggest that age-related improvements in interference inhibition stem primarily from enhanced cognitive conflict resolution, beyond general improvements in processing speed. From a neural perspective, age-related changes in drift rate likely reflect maturation of the prefrontal–anterior cingulate circuit. The anterior cingulate cortex monitors conflict, while the dorsolateral prefrontal cortex implements resolution strategies; structural and functional changes in this circuit are associated with enhanced conflict processing efficiency (Botvinick et al., 2001; Isherwood et al., 2023; Rueda et al., 2004). The more protracted developmental trajectory of interference inhibition may reflect the greater complexity of these neural circuits, which require extended periods to achieve full functional integration.

In contrast, for response inhibition, modeling analysis of the *Go/No-Go* task revealed that age-related improvements were primarily associated with systematic adjustments of starting point bias (z). The *Go/No-Go* task's asymmetric structure—high *Go*, low *No-Go* proportions—makes starting point bias a particularly important strategic variable. Unlike interference inhibition, response inhibition does not involve complex conflict resolution, so age-related changes manifest primarily in strategic aspects of decision-making rather than in evidence extraction efficiency. Behavioral differences traditionally interpreted as "inhibitory control ability" can be reconceptualized as individual differences in starting point bias toward the "*Go*" boundary (Forstmann et al., 2016). Age-related changes in starting point bias suggest that, with maturation, children and adolescents adopt increasingly flexible decision strategies when facing prepotent responses. This enhanced strategic adjustment ability likely reflects improvements in prefrontal-mediated top–down control (Miller & Cohen, 2001), enabling individuals to calibrate response tendencies according to task demands.

Despite these task-specific computational patterns, we identified an important shared foundation: nondecision time (t) exhibited significant age-related reductions in both tasks, consistent with shared processing components observed across inhibitory tasks (Isherwood et al., 2023; Palmer et al., 2025). Nondecision time encompasses processing outside the core decision process, including stimulus encoding, response preparation, and motor execution (Forstmann et al., 2016). This finding aligns with cognitive developmental theory positing that processing speed supports higher-level cognitive functions (Kail & Salthouse, 1994). Supporting this interpretation, Löffler et al. (2024) demonstrated that, when controlling for task-general processing components, task-specific EF differences were substantially reduced. With maturation, improvements in processing speed are associated with developmental changes throughout the central nervous system, contributing to performance gains across diverse cognitive domains (Kail, 2000).

Notably, our age-stratified analyses further revealed that computational dissociation between interference and response inhibition emerges gradually across development rather than representing a fixed property of the cognitive system. While prior studies have documented that EFs become increasingly differentiated with age (Brydges et al., 2014; Lee et al., 2013; Wiebe et al., 2011), the underlying computational mechanisms have remained largely unspecified. The present findings offer initial evidence that this differentiation involves a gradual transition from domain-general to task-specific decision-making processes. Specifically, early childhood was characterized by reliance on decision boundary (a) in both tasks, consistent with Wiebe et al.'s (2011) unity model of early EF and with Munakata et al.'s (2011) suggestion that young children rely more heavily on reactive control. This domain-general pattern gives way during middle childhood and early adolescence to more task-appropriate strategies, paralleling the transition from reactive to proactive control (Chatham et al., 2009; Czernochowski et al., 2010). By late adolescence, the dissociation is fully established, with drift rate (v) and starting point bias (z) serving as the primary task-discriminating features—broadly consistent with the differentiated factor

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structure observed in adult samples (Miyake et al., 2000). At the neural level, this progressive computational differentiation aligns with neuroimaging evidence showing that prefrontal activation patterns shift from diffuse to focal across development (Brydges et al., 2013; Durston et al., 2006), and that functional connectivity within task-specific networks strengthens throughout adolescence (Luna et al., 2015). The computational transition from threshold-based to parameter-specific control may thus reflect the gradual refinement and specialization of underlying neural circuits. Together, these findings contribute to an emerging developmental computational neuroscience perspective (Van Den Bos & Hertwig, 2017) suggesting that the mature computational architecture of inhibitory control emerges through prolonged development rather than reflecting an innate organizational principle. Interestingly, this computational differentiation coexists with recent neuroimaging evidence indicating that adults show greater neural similarity between interference inhibition and response-inhibition tasks than do adolescents (Palmer et al., 2025). This pattern suggests that mature inhibitory control may involve task-specific computational optimization operating within increasingly integrated neural networks.

A decision-computational framework for inhibitory control development

Building on our findings and recent theoretical developments (Löffler et al., 2024; Rey-Mermet et al., 2018; Simpson & Carroll, 2019), we propose a decision-computational framework that reconceptualizes inhibitory control development as systematic optimization of decision parameters rather than linear growth in a unitary ability (Figure 5). The framework's core proposition is that different inhibitory subcomponents optimize distinct computational parameters—interference inhibition is associated with enhanced evidence accumulation (*v*), while response inhibition involves calibration of starting point bias (*z*)—yet share common foundations in processing speed (*t*) and, to a lesser extent, response caution (*a*). This architecture explains the moderate

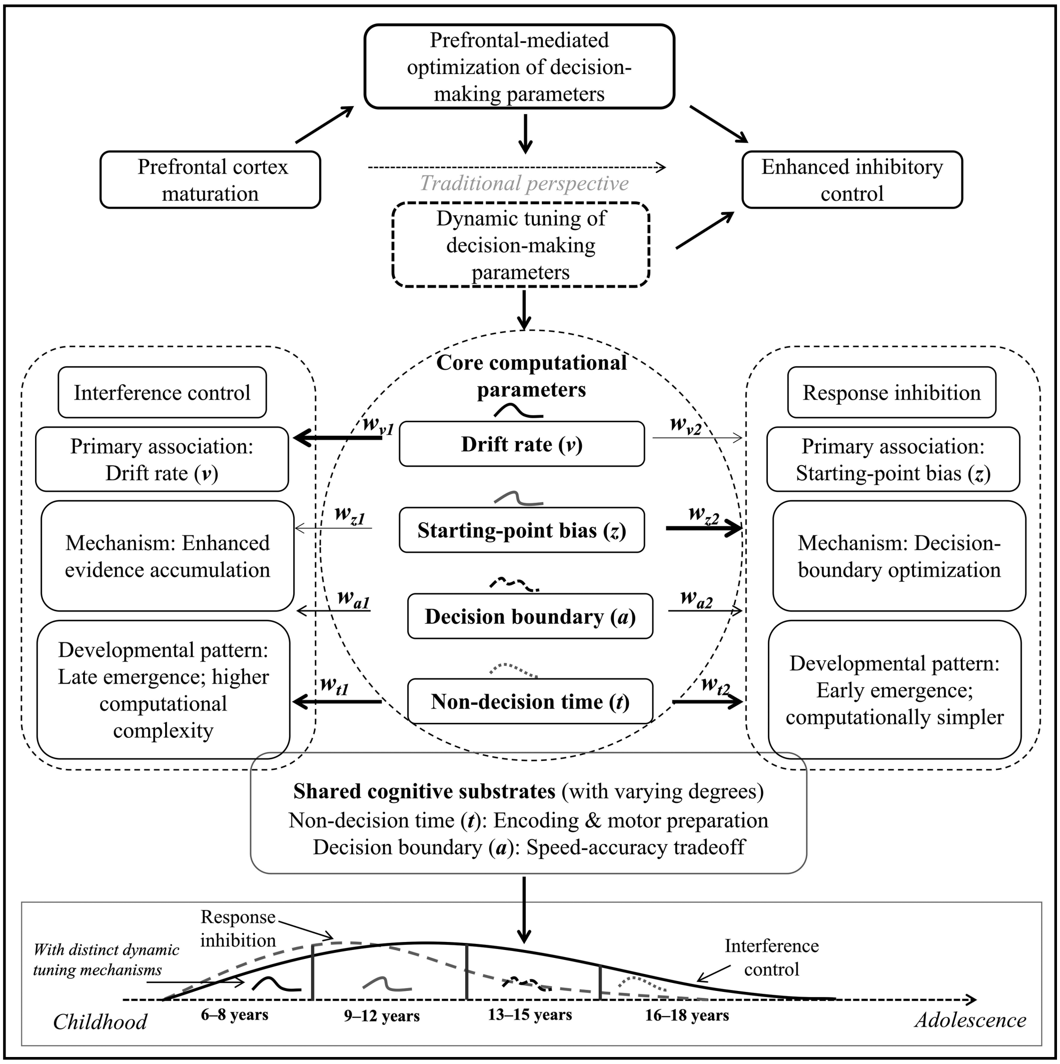

Figure 5 A decision-computational framework for inhibitory control development. The framework illustrates task-specific optimization of decision parameters (drift rate for interference inhibition, starting point bias for response inhibition) with shared foundations in processing speed and response caution. Arrow weights (*W*) represent the relative contribution of each parameter to developmental change in each inhibitory control component. Subscripts 1 and 2 denote the relative weight of each parameter in interference inhibition and response inhibition, respectively. *a* = decision boundary; *t* = nondecision time; *v* = drift rate; *w* = weight; *z* = starting point bias.

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intercorrelations consistently observed between inhibitory tasks (Friedman & Miyake, 2004) while accounting for their distinct developmental trajectories. Critically, this computational architecture is not fixed but emerges progressively across development, transitioning from domain-general threshold-based control in early childhood to task-specific parameter optimization by late adolescence. This perspective also reframes individual differences: children may achieve similar behavioral outcomes through distinct computational routes—some primarily enhancing drift rates (processing efficiency), others adjusting thresholds (response conservatism)—suggesting that equivalent performance can mask qualitatively different underlying strategies (Ratcliff et al., 2012). This framework extends unity/diversity models (Friedman & Miyake, 2017; Miyake & Friedman, 2012) by specifying the computational mechanisms underlying both shared and distinct aspects of inhibitory control, and by demonstrating that computational specialization itself represents a developmental achievement rather than an innate organizational principle.

Limitations and future directions

Although this study provides novel evidence for understanding developmental patterns of inhibitory control, several limitations warrant consideration. First, we did not systematically examine how demographic factors might moderate the observed developmental patterns. Given evidence that socioeconomic status influences EF development through effects on prefrontal cortex structure and function (Hackman & Farah, 2009; Noble et al., 2015), future research should investigate whether variables such as socioeconomic status, parental education, or family structure moderate nonlinear trajectories or computational parameter development. Second, our developmental coverage is constrained in both duration and age range. Although the 6-month follow-up provided preliminary validation of trajectory stability, understanding enduring changes in computational parameters requires longer observation periods. Additionally, we did not include children younger than 6 years, a period when inhibitory control undergoes rapid initial development (Berger et al., 2022; Garon et al., 2008; Kochanska et al., 2000), limiting our ability to determine when interference and response inhibition first diverge. Third, broader construct and task coverage is needed to establish generalizability. We focused on interference and response inhibition using a single paradigm per construct, and we did not examine other inhibitory components (e.g., cognitive inhibition, oculomotor inhibition; Friedman & Miyake, 2004; Nigg, 2000) or incorporate temperament-based assessments (Rothbart et al., 2003). Whether the distinct parameter patterns generalize remains an open question, as different tasks may elicit distinct cognitive strategies (Rey-Mermet et al., 2018) and even similar paradigms may recruit different neural mechanisms (Raud et al., 2020). Future research should employ multiple paradigms per construct and examine cross-cultural measurement equivalence. Finally, a related consideration involves developmental measurement equivalence—whether identical tasks engage qualitatively different processes at different ages (Petersen et al., 2016; Simpson & Carroll, 2019). We addressed this through measurement invariance testing and HDDM analyses, which confirmed that the same computational architecture captured behavioral patterns across ages 6–18 years, with changes manifesting as quantitative parameter shifts rather than qualitative model changes. Nevertheless, future research could strengthen these conclusions through converging evidence from experimental manipulations or neuroimaging.

Conclusions and implications

This study provides initial computational evidence for both unity and diversity in inhibitory control development across childhood and adolescence. By integrating flexible developmental modeling with computational approaches, we found that interference and response inhibition follow distinct developmental trajectories associated with different computational mechanisms—enhanced evidence accumulation versus strategic threshold adjustments, respectively—while sharing improvements in processing speed that may reflect common developmental foundations. Importantly, this computational dissociation emerges gradually across development, with young children showing greater reliance on domain-general strategies before transitioning to more task-specific mechanisms by late adolescence. These findings contribute to a decision-computational framework that advances understanding of inhibitory control from a descriptive construct toward a more mechanistically grounded model of cognitive development.

Our findings carry potential implications for educational and clinical practice. In educational settings, the developmental asynchrony between response inhibition and interference inhibition suggests that age-specific instructional approaches may be beneficial. As response inhibition stabilizes earlier, curricula for older adolescents might place greater emphasis on activities targeting interference inhibition—challenging students to reconcile competing information and maintain goal-relevant focus. In clinical contexts, computational parameter analysis could enable more precise identification of specific deficits. For instance, some children with attention-deficit/hyperactivity disorder may exhibit difficulties primarily in evidence accumulation (drift rate) rather than response strategy (starting point bias), potentially pointing to different intervention targets. This parameter-based approach could provide more nuanced diagnostic information and guide more targeted interventions, though further research is needed to establish the clinical utility of these computational distinctions.

Supplementary material

Supplementary material is available at *Child Development* online.

Data availability

The data necessary to reproduce the analyses presented here are available from the corresponding author (Xin Zhao) upon reasonable request. The analytic code necessary to reproduce the analyses presented here is publicly accessible at https://github.com/tyzhang98/inhibitory-control-dev-cogmodel-code. The materials necessary to attempt to replicate the findings presented here are available from the corresponding author (Xin Zhao) upon reasonable request. The design and analysis plans for this study were not preregistered.

Author contributions

Tongyi Zhang (Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing [lead], Formal analysis [equal]), Yajie Gong

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(Conceptualization, Writing—original draft, Writing—review & editing [supporting], Formal analysis [equal]), and Xin Zhao (Conceptualization, Funding acquisition, Project administration, Resources, Supervision [lead], Writing—original draft, Writing—review & editing [equal].)

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Conflicts of interest

None declared.

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